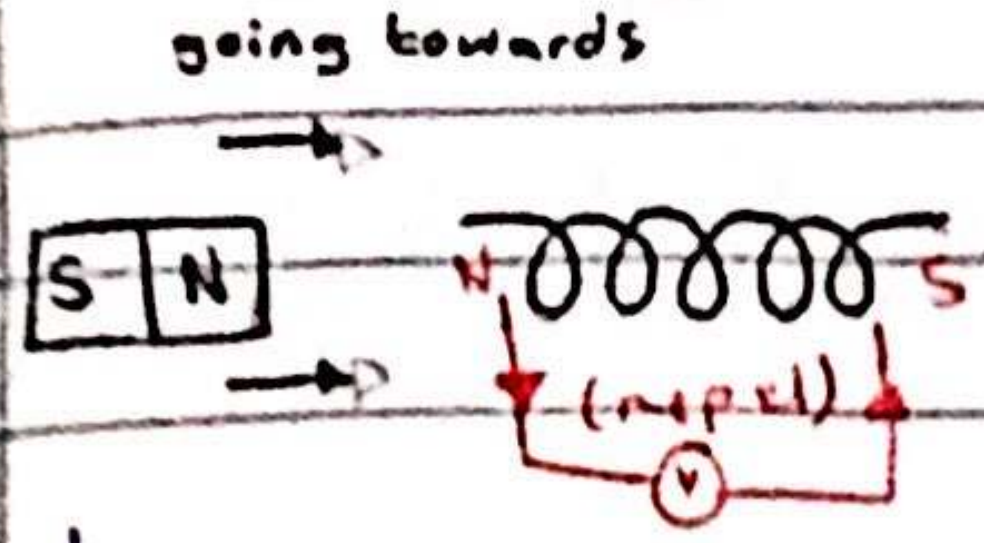
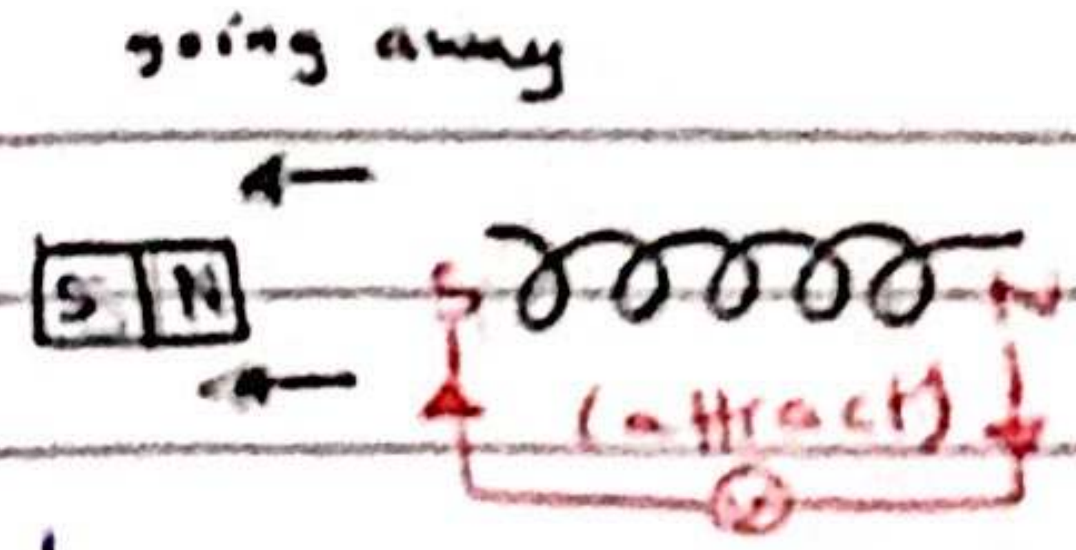
Electromagnetic induction:-

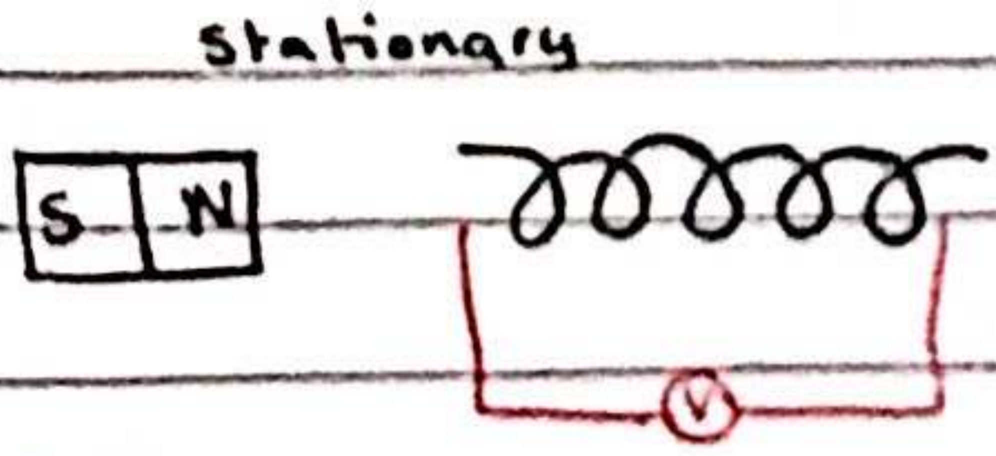
- a conductor moving across a magnetic field or a changing magnetic field linking with a conductor can induce an e.m.f in the conductor
- direction of an induced e.m.f opposes the change causing it.





↳ here the voltmeter's needle will go to the right

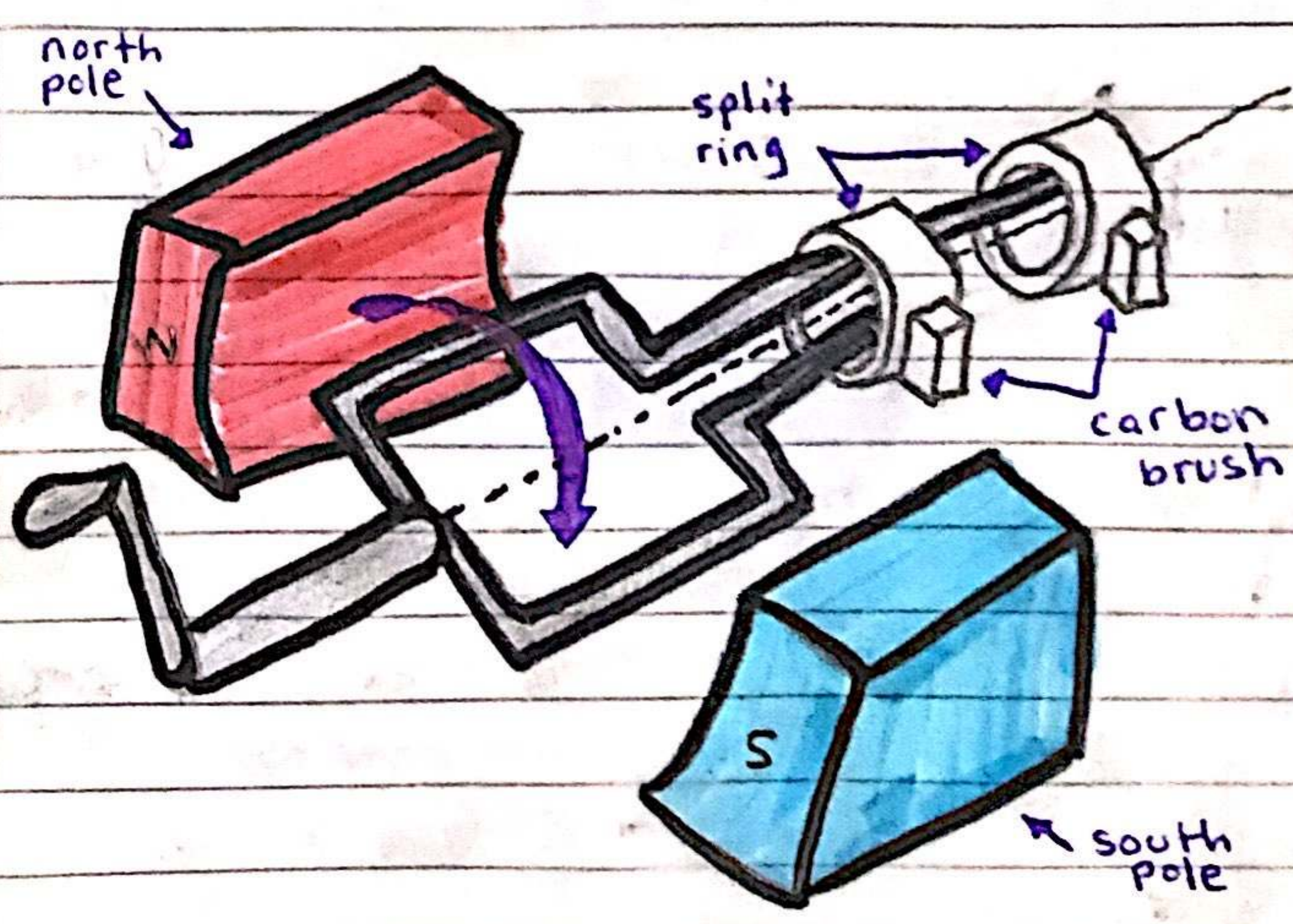
↳ here the voltmeter's needle will go to the left

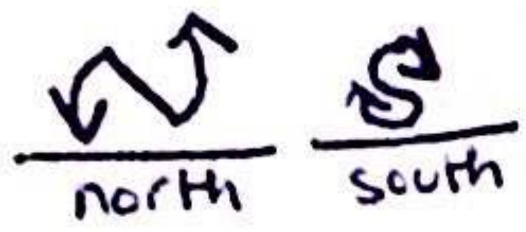


↳ here the voltmeter's needle won't move

Factors:-
- moving the magnet/wire faster
- increase the number of loops of wire
- stronger magnet

A.C generator :-
- makes a.c current using the fundamentals of electromagnetic induction





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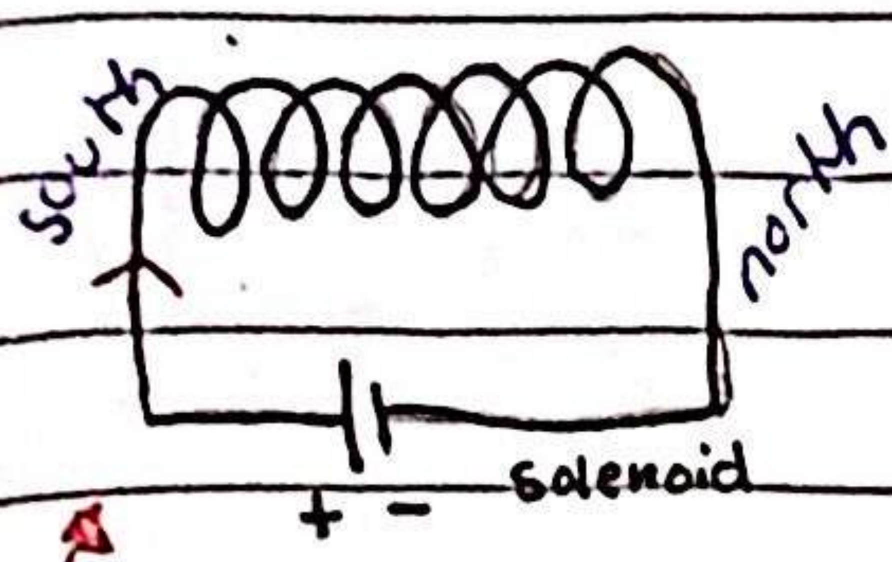
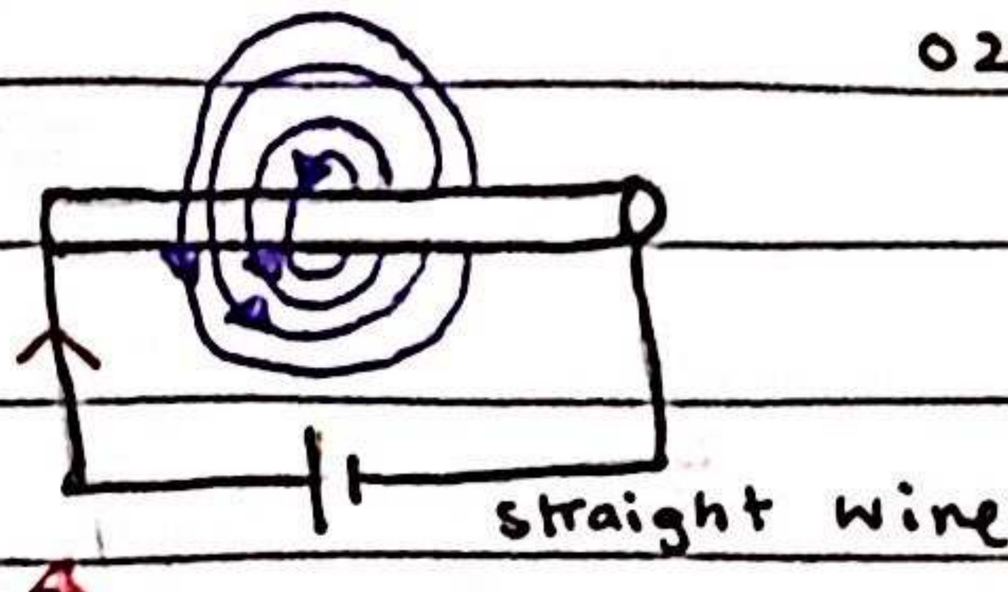




front view

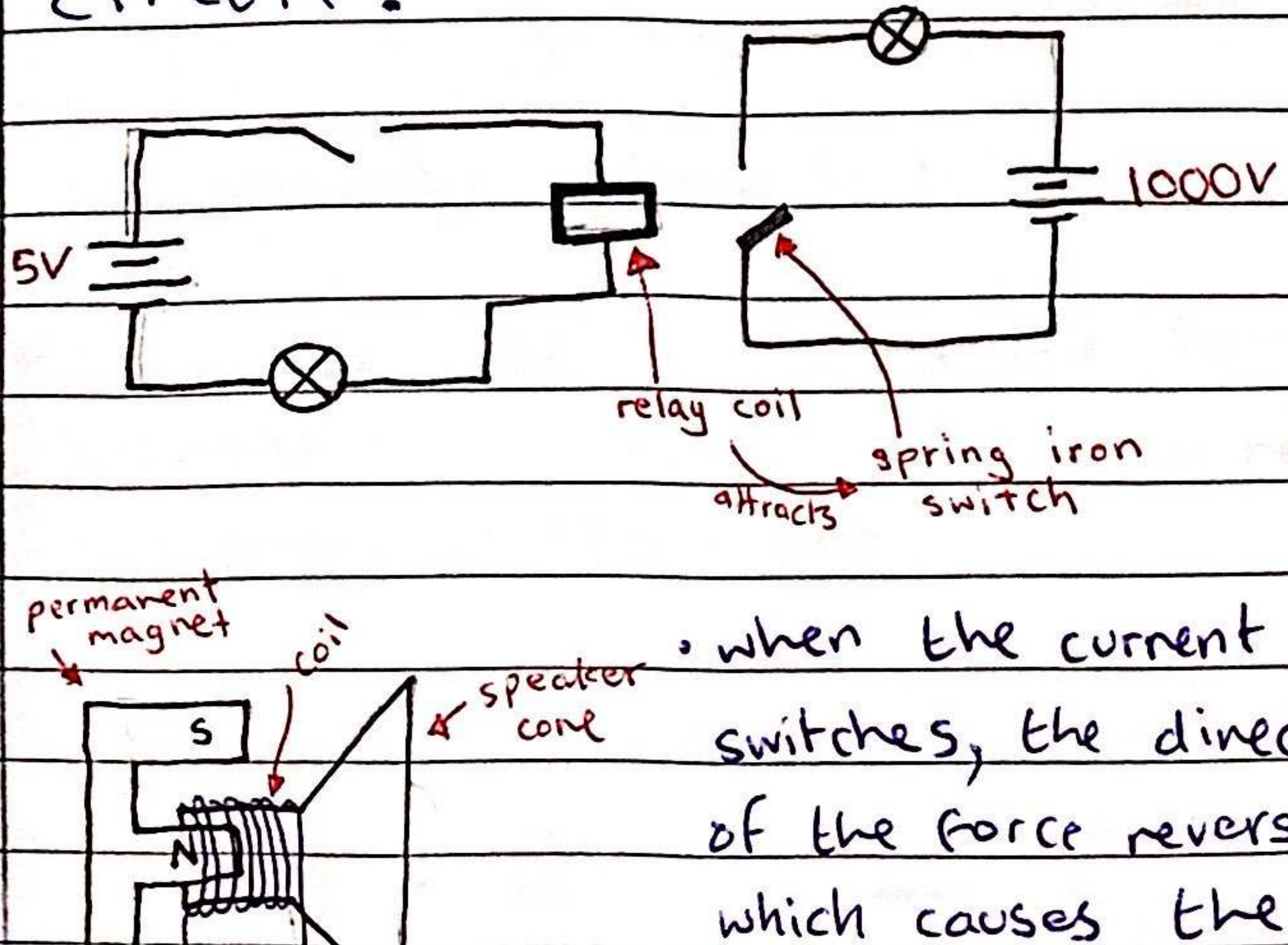
right hand rule

the direction of the mag. field is the direction of the FORCE on a NORTH POLE at that point.

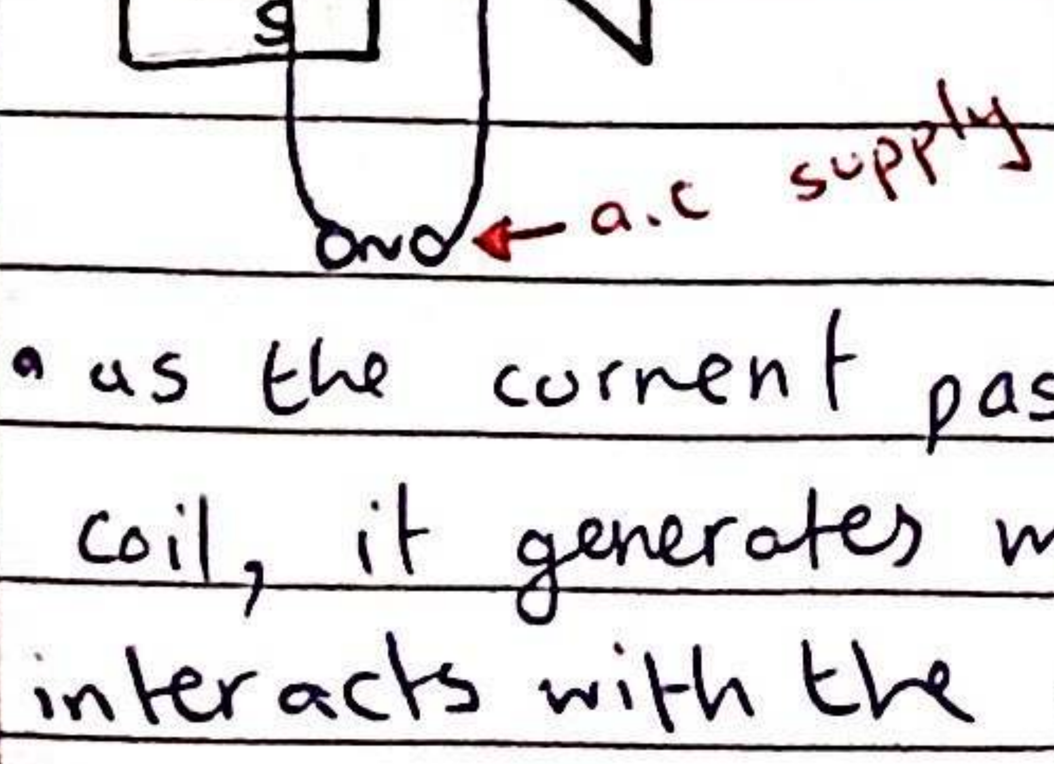
the closer the mag. field lines, stronger the mag. field

Relays:-

- allows a low voltage/current circuit to turn on a high voltage/current circuit.



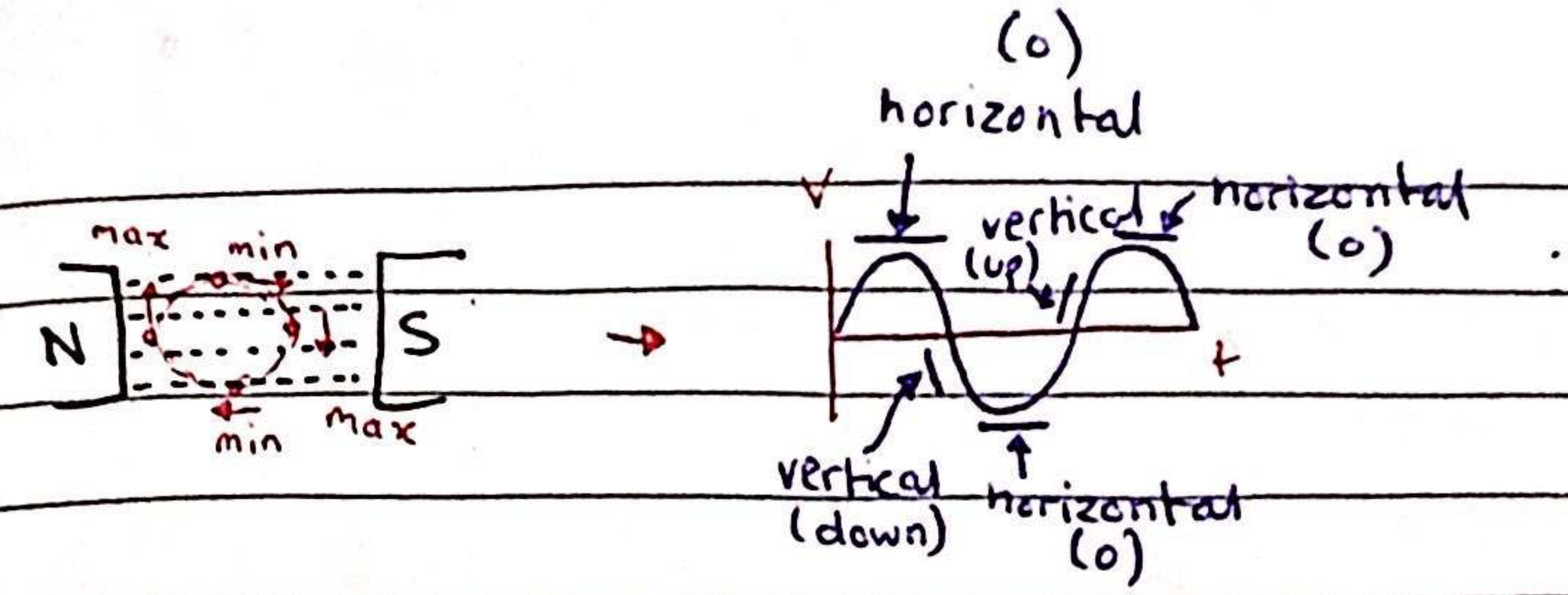
- the coil, once activated, attracts the iron switch therefore turning on the high voltage circuit.



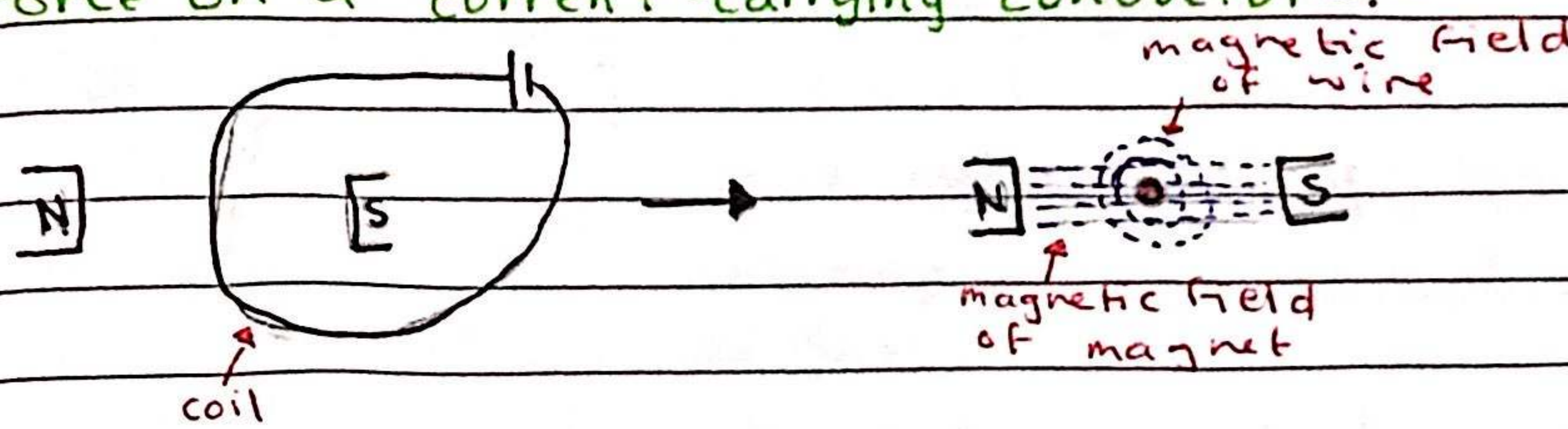
- when the current switches, the direction of the force reverses which causes the cone to move in & out

* bigger the magnitude, stronger the mag. field
* changing the direction will change the polarity.

- as the current passes through the coil, it generates mag. field which interacts with the permanent magnet

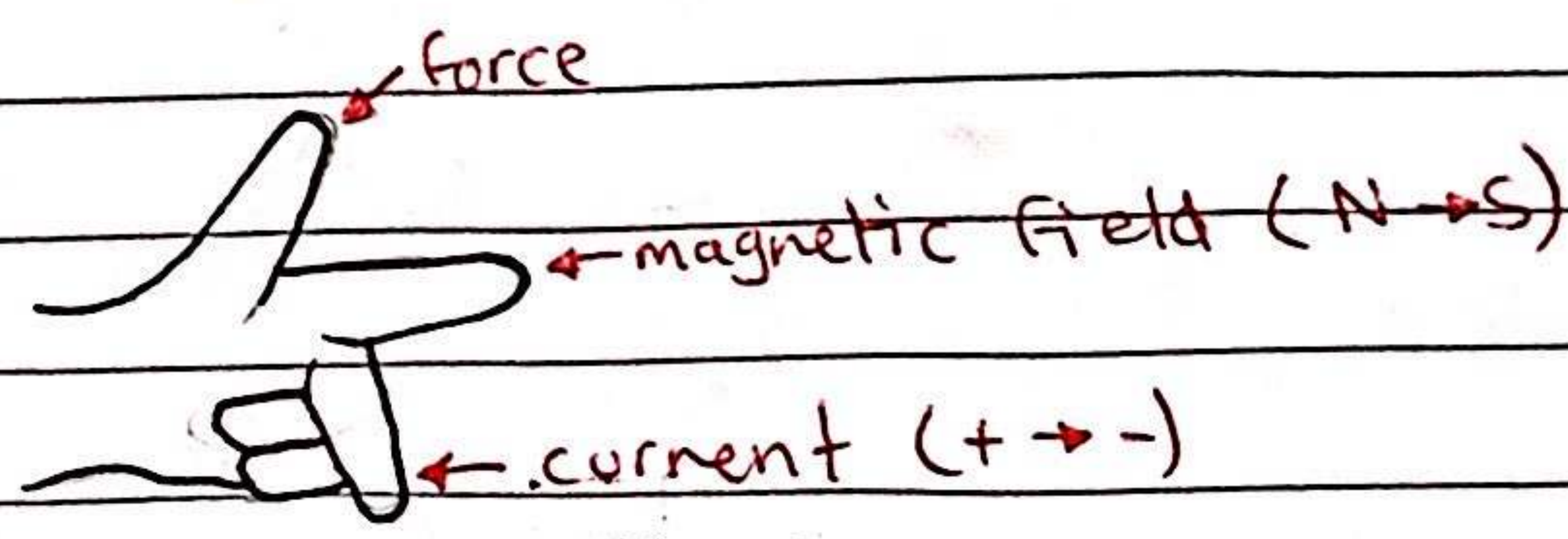


Force on a current-carrying conductor -:-



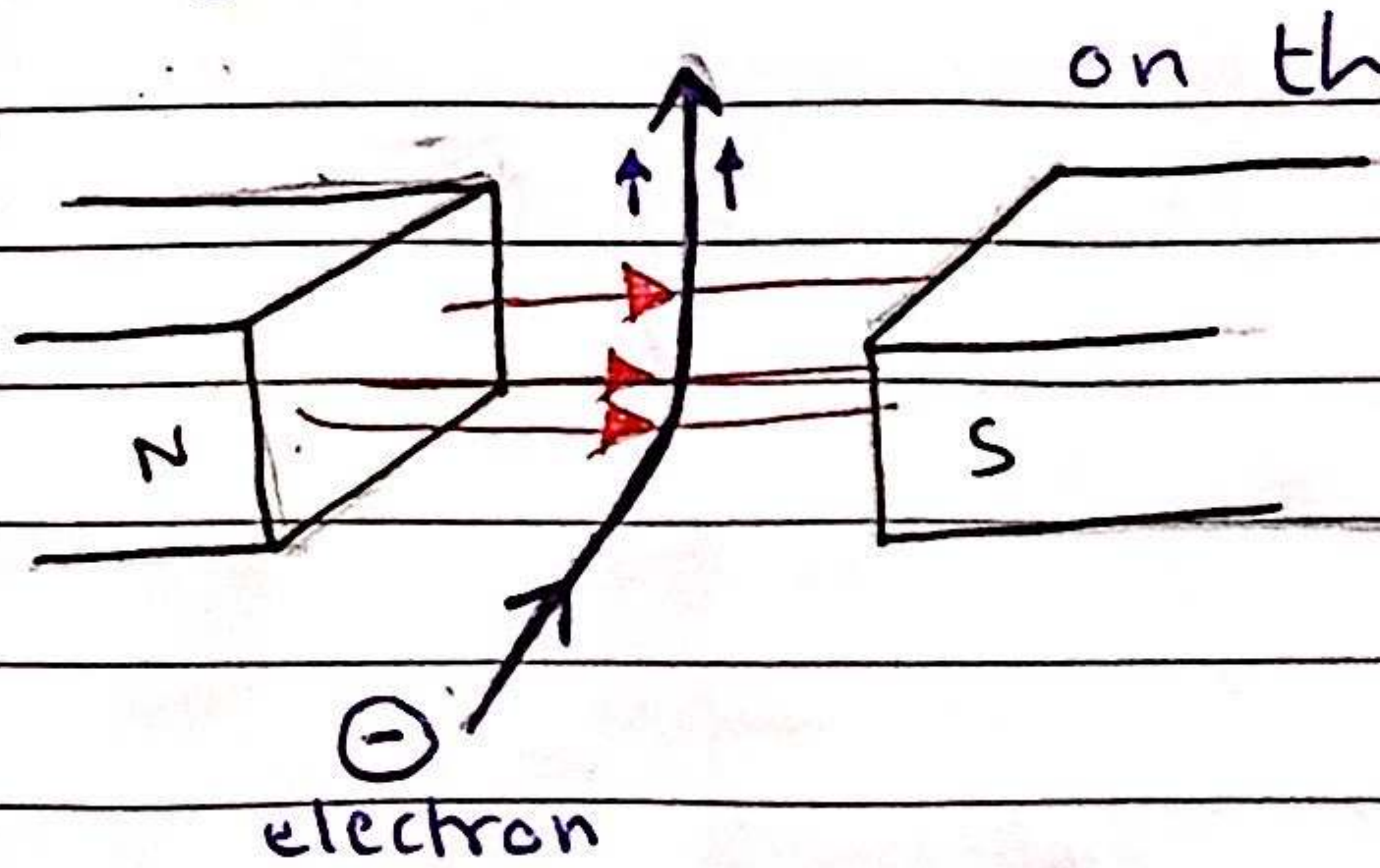
* the mag. field of the magnet & the wire interact which creates a up or down force

↳ Fleming's left hand rule

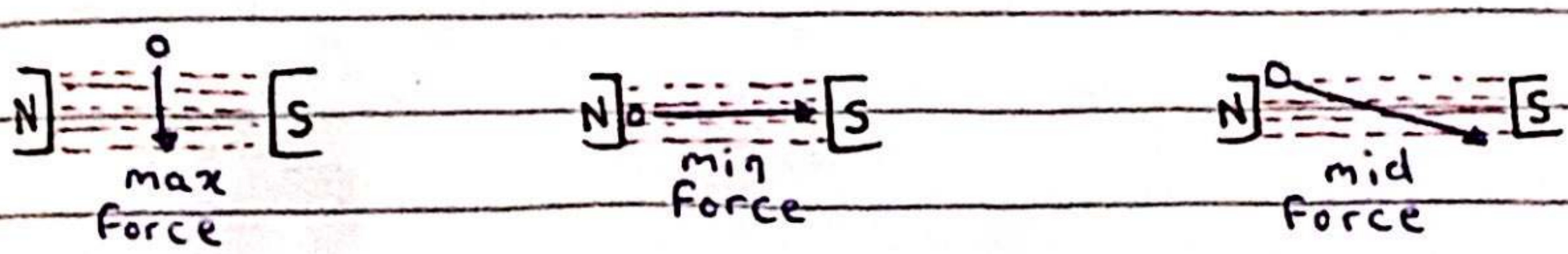


- to reverse the direction of the force, you can change the direction of the current or the direction of the field.

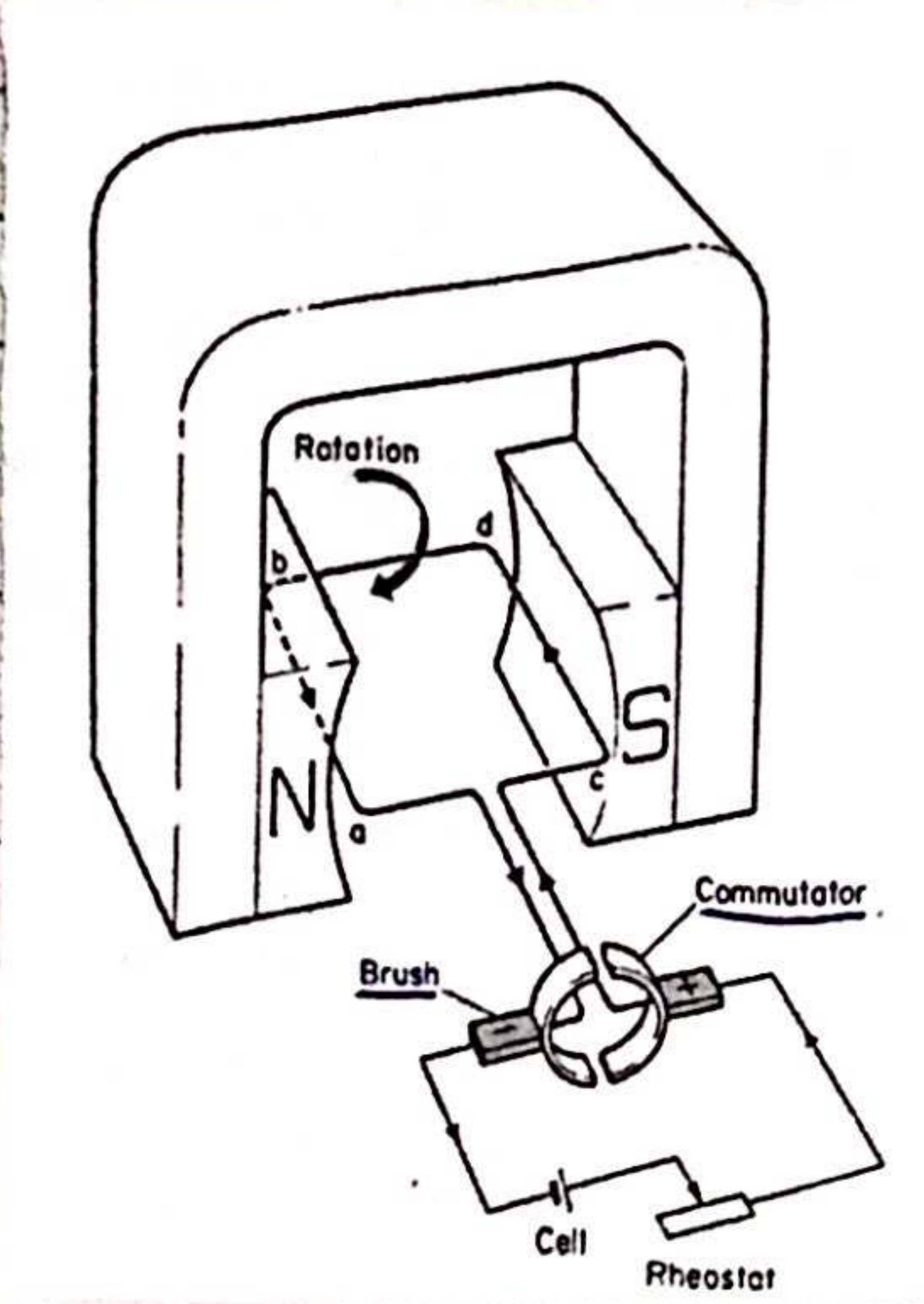
- charged particle → the field will exert a force on the particle, causing it to deflect



* for an electron, the 'current' finger points in the opposite direction



D.C Motor :-



split ring (commutator):
reverses the current every half-cycle to keep the coil rotating in the same direction

brushes:
transmit current from battery to the coil w/o breaking them and helps reverse the current

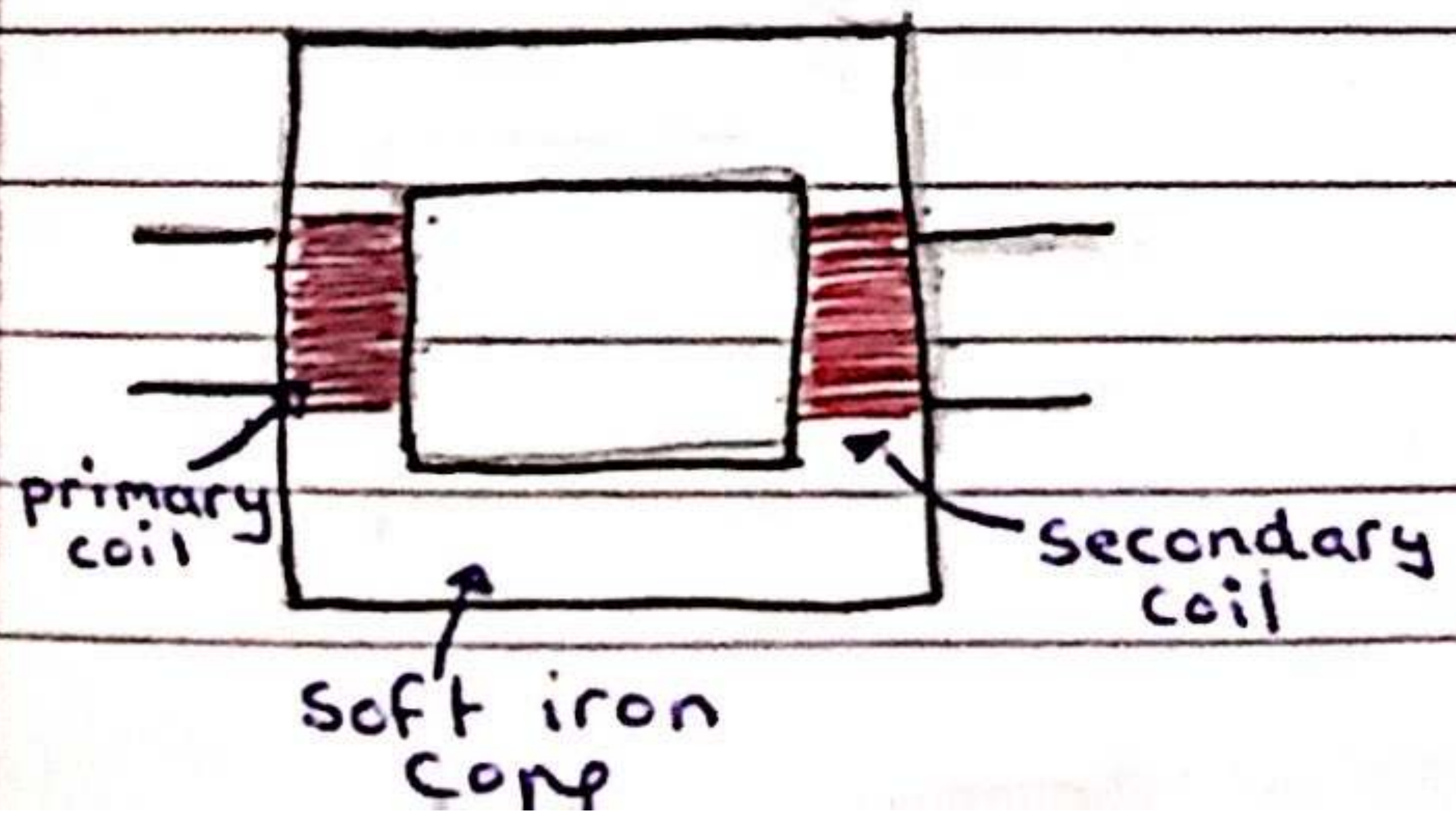
* a current-carrying coil in a magnetic field may experience a turning effect

Factors affecting the turning effect:-

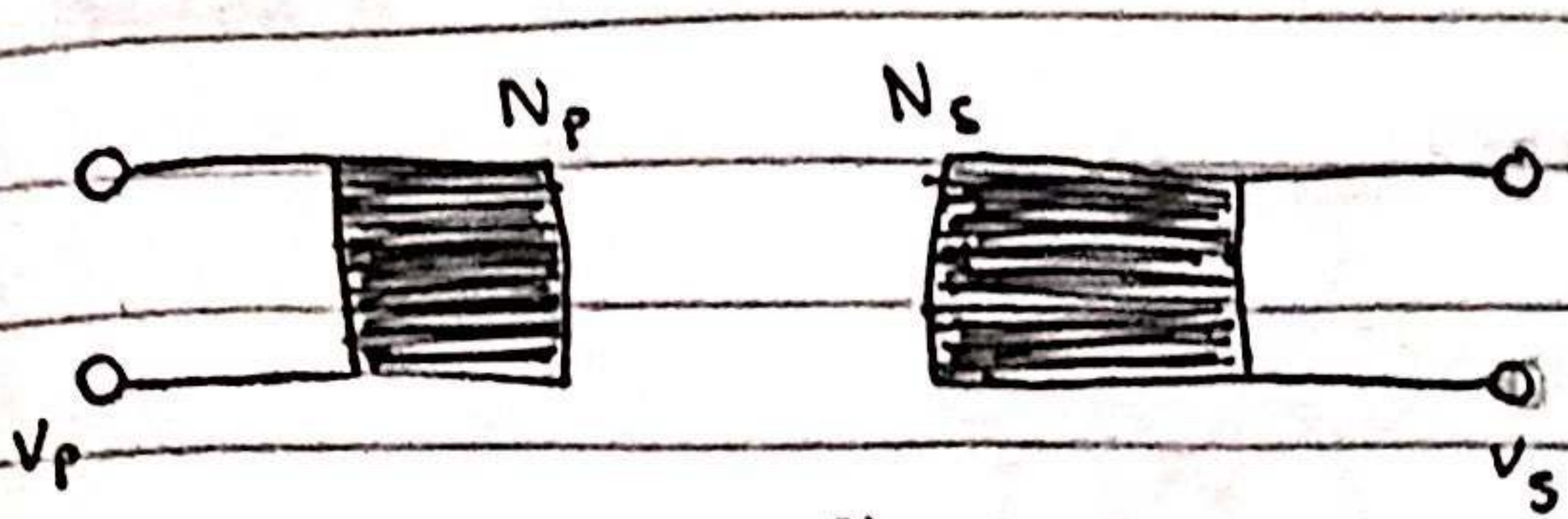
- the number of turns on the coil
- the current
- the strength of the magnetic field

Transformer :-

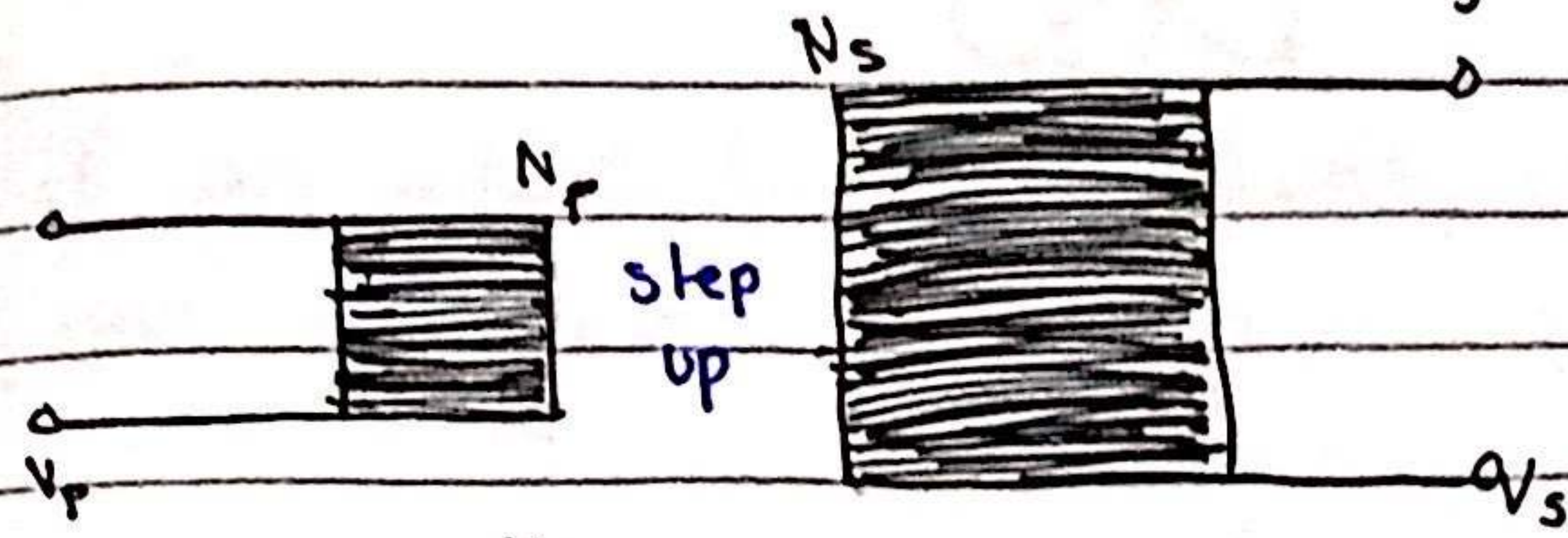
- device that increases/decreases the voltage of an a.c supply w/o losing power.



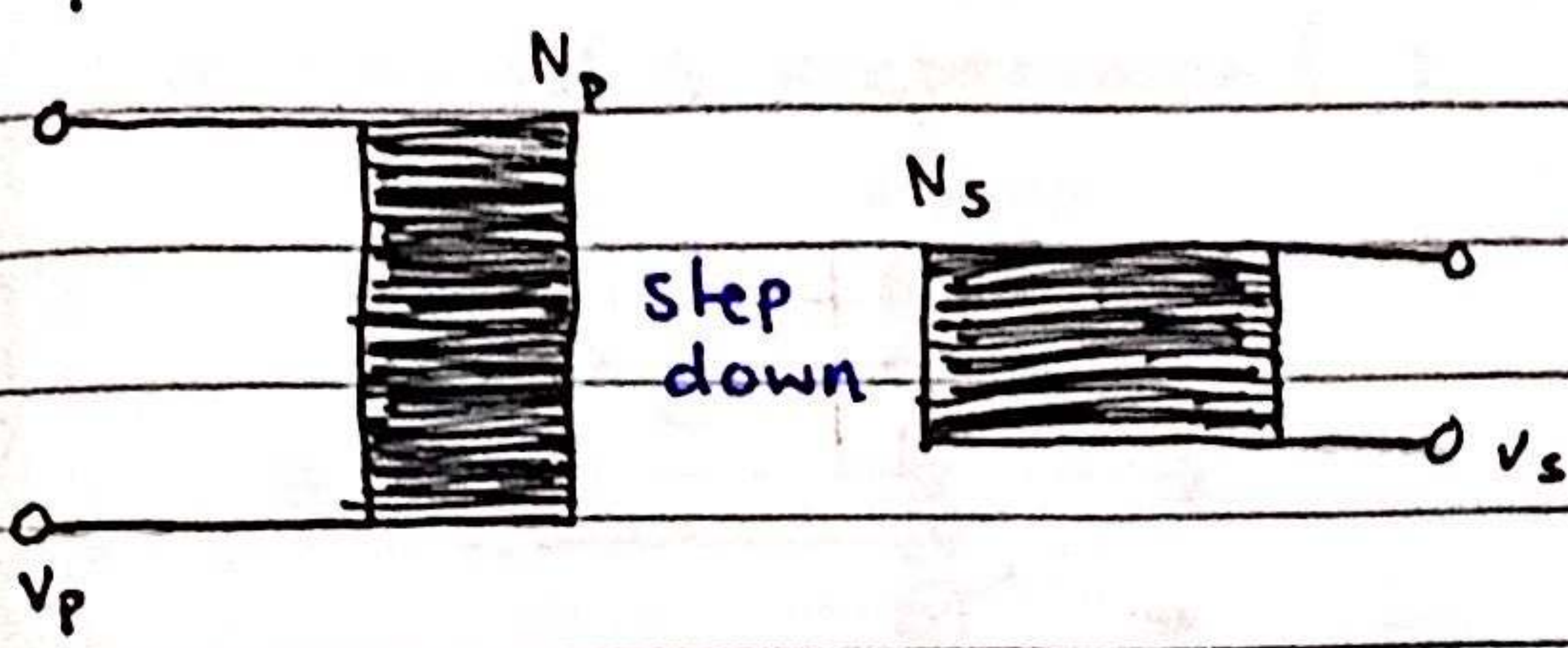
* iron is used because it is easily magnetised



= 100 V (input) → 100 V (output)

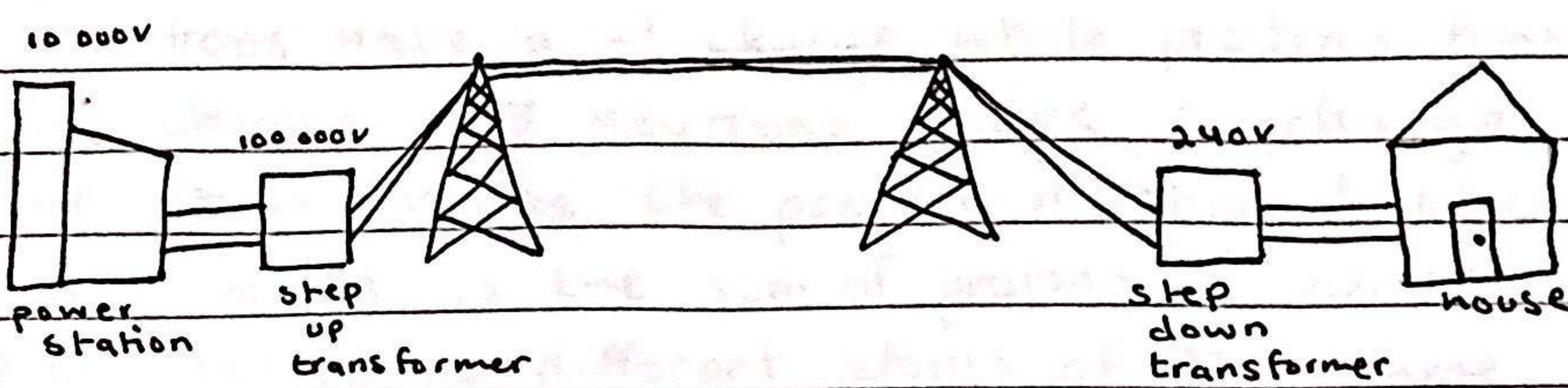


= 100V (input) → >100V (output)



= 100 V (input) → <100V (output)

* $\frac{V_p}{V_s} = \frac{N_p}{N_s}$
  - ↳ this shows that the ratio of the voltage across the pri./sec. coils of a transformer is equal.
* $I_p V_p = I_s V_s$
  - ↳ if a transformer is 100% efficient
* $P = I^2 R$
  - ↳ the amount of power lost in the wire



advantages

+ the wires are really long, so high current would damage the wires/heat them up thus a loss of energy therefore the voltage can be increased so the current stays low.